Marmara University
Engineering Faculty
Electrical and Electronics Engineering
EE4077: Fundamentals of Machine Learning
Midterm Exam (Spring, 2024)

1. Consider the following data set with three binary features X_1, X_2, X_3, and a binary label Y.

X_1	X_2	X_3	Y
1	0	1	0
0	0	1	0
1	1	0	0
0	0	0	1
1	1	1	1
1	0	0	1

   (a) [10 pt] We wish to fit a Naive Bayes model to the data $P(Y|X_1,\ldots,X_n) \propto P(Y)\prod_{i=1}^{n} P(X_i|Y)$. Calculate the probability $P[Y = 0|X = (0,1,1)]$.
   (b) [15 pt] Apply Laplace smoothing with $k = 1$. Calculate $P[Y = 1|X = (0,1,0)]$.

2. [25 pt] Suppose you are an avid BTS (a South Korean boy band that formed in 2010 and debuted in 2013) stan who monitors the social media accounts of each of the members. Suppose you wish to find the probability that a BTS member will post at any time of day. Over the three days period, you look on Instagram and find the following number of new posts: x = [3,4,1]. A fellow stan tells you that this comes from a Poisson distribution.

$$p(x|\theta) = \frac{e^{-\theta}\theta^x}{x!}$$

   where θ is an unknown parameter that we want to estimate. You are told that θ has the following probability density function (PDF)

$$p(\theta) = \frac{1}{4}\theta e^{-\frac{\theta}{2}},\ \theta > 0$$

   Calculate maximum a posteriori probability (MAP) estimate $\hat{\theta}_{MAP}$.

3. In this problem, you will see how to learn a decision tree. Consider the following data set with three features X_1, X_2, X_3, and label Y.
   (a) [10 pt] Suppose calculate the mutual information for each feature (X_1, X_2, X_3) to determine the root node for a Decision Tree trained on the data. What feature should be split on at the root node?
   (b) [10 pt] Calculate what the next split should be.
   (c) [5 pt] Draw the resulting tree.

Outlook (X_1)	Temperature (X_2)	Humidity (X_3)	Play Tennis? (Y)
sunny	hot	high	no
overcast	hot	high	yes
rain	mild	high	yes
rain	cool	normal	yes
sunny	mild	high	no
sunny	mild	normal	yes
rain	mild	normal	yes
overcast	hot	normal	yes

4. Answer following true/false questions. If you answer true, provide a short justification, if false explain why or provide a small counterexample. Answers without justification will get no credit.

   (a) [5 pt] **True/False.** If the data is not linearly separable, then there is no solution to the hard-margin Support Vector machine (SVM).
   (b) [5 pt] **True/False.** Logistic Regression cannot be used for classification.
   (c) [5 pt] **True/False.** In stochastic gradient descent, we take steps in the exact direction of the gradient vector.
   (d) [5 pt] **True/False.** It is not a good machine learning practice to use the test set to help adjust the hyperparameters of your learning algorithm.
   (e) [5 pt] **True/False.** Cross validation will guarantee that our model does not overfit.

Directions:
1. Closed book, closed notes
2. The exam is 75 minutes.
3. Justify your answers.